Intermediate Elective

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Packages

```r
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(lubridate)
library(dplyr)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```r
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```r
# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

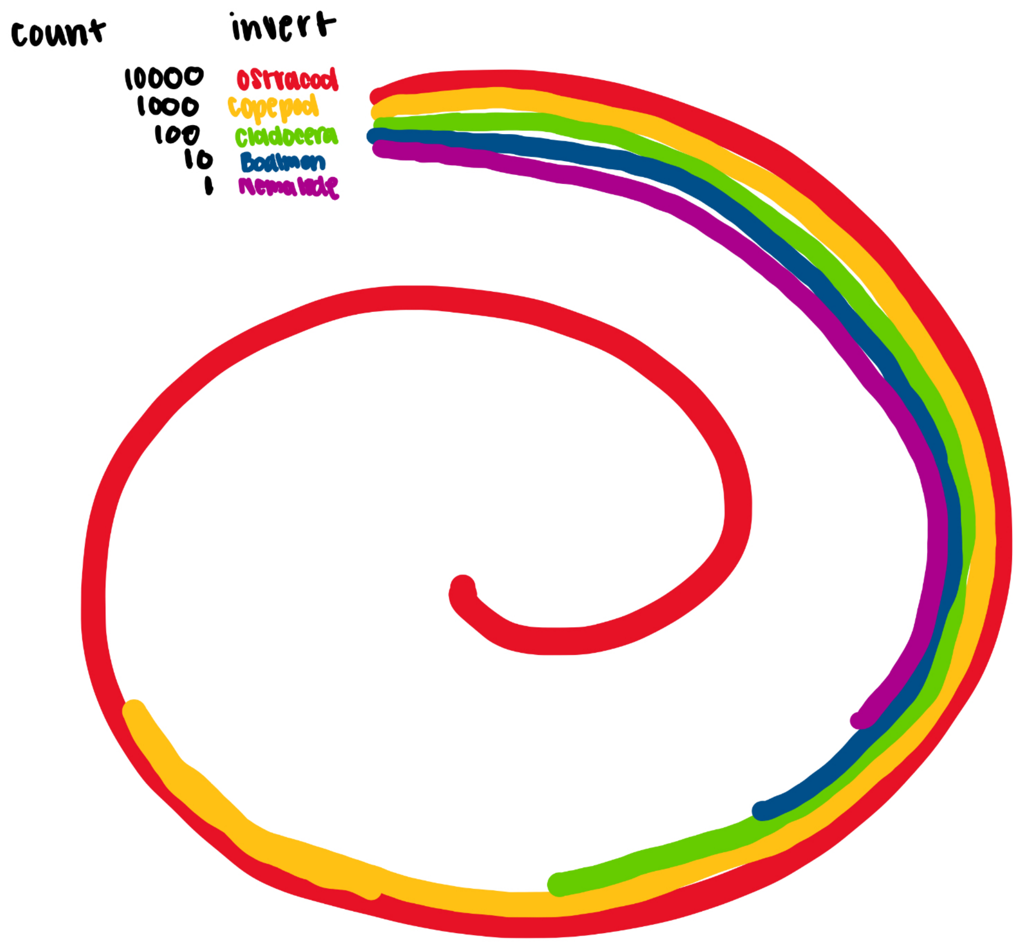
1. Explain your inspiration

- I picked the spiral looking graphic labled 2021_06_dubois_data graphic by Ijeamaka Anyenes.
- This graphic makes sense for my dataset of choice because it is able to display differences in abundance in a fun yet clear and simple way.
- The visualization will show the 5 most abundant inverts during 2024. Each line will represent a different invert and have its own color while the length of the line represents abundance.

2. Plan your figure

- Final visual goal: compare the abundances of the most common inverts found at ncos in 2024

```r
knitr:: include_graphics(here("code","IMG_2970.jpg"))
```



#3. Code your figure

Create data frame

```
invert_2024 <- aq_ins |>
  clean_names() |>
  rename(date = date_on_vial) |>
  filter(year(date) == 2024) |>
  select(ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
```

```
        nematode) |>
  pivot_longer(cols = ostracod:nematode,
               names_to = "taxon",
               values_to = "abundance") |>
  mutate(
    abundance = replace_na(abundance, 0),
    taxon = str_to_title(str_replace_all(taxon, "_", " "))
  ) |>
  group_by(taxon) |>
  summarise(total_abundance = sum(abundance), .groups = "drop")
```